Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Williams, Gordon H. M.D.		
Project Title:	Effects of Microgravity on Renal, Endocrine and Volume Regulatory Function		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gwilliams@partners.org	**Fax:**	FY 617-732-5764
PI Organization Type:	UNIVERSITY	**Phone:**	617-525-7490
Organization Name:	Brigham and Women's Hospital		
PI Address 1:	221 Longwood Avenue		
PI Address 2:			
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Original end date was 3/31/2008.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Richard (Massachusetts Institute of Technology) Grenon, Marlene (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-CA00406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Orthostatic intolerance remains an operational problem following spaceflight. In addition, there is growing evidence that cardiac dysrhythmias may pose a threat to the health of space travelers. At present, no effective countermeasures exist for orthostatic intolerance and for the probable increased risk of ventricular dysrhythmias. In the last six years, we have addressed the subject of renal, endocrine and volume-regulating function during head-down tilt bed rest. We have found that simulated microgravity induces changes in the volume-regulating systems and electrolyte excretion, and leads to changes in cardiac electrical stability. We have also found that using our tilt-test protocol, the pre-bed rest tilt-test tolerance was predictive of post-bed rest tilt-test tolerance. We propose now to conduct two double-blinded randomized trials to: 1) Investigate the use of a pre-bed rest tilt-test as a means to screen countermeasures against orthostatic intolerance, and investigate the influence of these countermeasures on the renal and endocrine responses to orthostatic stress. 2) Investigate the effectiveness of individualized countermeasures identified during a pre-bed rest tilt-test in reducing post-bed rest orthostatic intolerance, as well as their influence on the renal and endocrine responses to orthostatic stress. 3) Investigate the effects of different countermeasures in reducing cardiac electrical instability, and to study the effects of these countermeasures on baseline orthostatic tolerance. 4) Investigate the effects of different countermeasures in reducing cardiac electrical instability after simulated microgravity, and to study the effects of these countermeasures on orthostatic tolerance and on the renal and endocrine responses to simulated microgravity. As with our current grant, this one relates to a companion application “Effects of Microgravity on Cardiovascular Function” by Richard Cohen, M.D., Ph.D. These studies have implications for the treatment and prevention of maladaptive hemodynamic responses experienced by astronauts in flight and on return to Earth. They will provide an innovative way of testing “individualized countermeasures”. Finally, the results of these studies may help us better understand the pathophysiology of alterations in volume homeostatic mechanisms in cardiovascular diseases such as congestive heart failure, dysrhythmias and hypertension.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has implications for the treatment and prevention of maladaptive hemodynamic responses experienced by astronauts in flight and on return to Earth. It will increase our understanding of the mechanisms by which weightlessness changes volume and sodium homeostasis, and possibly cardiac electrical stability, thereby providing a starting point to develop appropriate countermeasures. Perhaps most importantly, it will broaden our database to include older individuals, a group that is well represented among our population of current and future space travelers. Finally, the results of these studies may further our understanding of the pathophysiology of alterations in volume homeostatic mechanisms in cardiovascular diseases such as congestive heart failure.
Task Progress:	We are enrolling subjects into the study.
Bibliography Type:	Description: (Last Updated: 01/22/2008)
Articles in Peer-reviewed Journals	ú Grenon SM, Hurwitz S, Xiao S, Sheynberg N, Judson B, Ramsdell CD, Kim C, Cohen RJ and Williams GH "The Effect of Sleep Restriction on Orthostatic Intolerance in Simulated Microgravity Journal of Applied Physiology" 1660-1666 , Jan-2005
Articles in Peer-reviewed Journals	ú Grenon SM, Hurwitz S, Xiao S, Sheynberg N, Ramsdell CD, Kim C, Cohen RJ and Williams GH "Re-Adaptation from Simulated Microgravity as a Stimulus for Improved Orthostatic Tolerance : Variables Implicated Journal of Investigative Medicine" Jan-2005
Articles in Peer-reviewed Journals	ú Xiao X, Grenon SM, Kim C, Sheynberg N, Williams GH, and Cohen RJ "Effects of Prolonged Bed Rest on Human Calf Hemodynamics and Orthostatic Intolerance " Jan-2005
Presentation	Aú Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Cohen RJ, Williams GH "The Effect of Sleep Restriction on Orthostatic Intolerance in the Simulated Microgravity Environment " Oct-2004
Presentation	Aú Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Cohen RJ, Williams GH "The Role of Individual Predisposition in Orthostatic Intolerance Before and After Simulated Microgravity " May-2004
Presentation	Aú Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Kim C, Williams GH, Cohen RJ "Simulated Microgravity Induces Microvolt T-Wave Alternans " Oct-2004
Presentation	Aú Grenon SM, Xiao X, Sheynberg N, Hurwitz S, Kim C, Williams GH, Cohen RJ "Simulated Microgravity Induces Microvolt T-Wave Alternans " Oct-2004
Presentation	S. R. Bheemreddy, SM Grenon, X. Xiao, S. Hurwitz, N. Sheynberg, C. Kim, RJ. Cohen, GH. Williams "Cardio-Endocrine, Renal Responses and Influence of Gender in Humans to Simulated Microgravity " Jan-2005